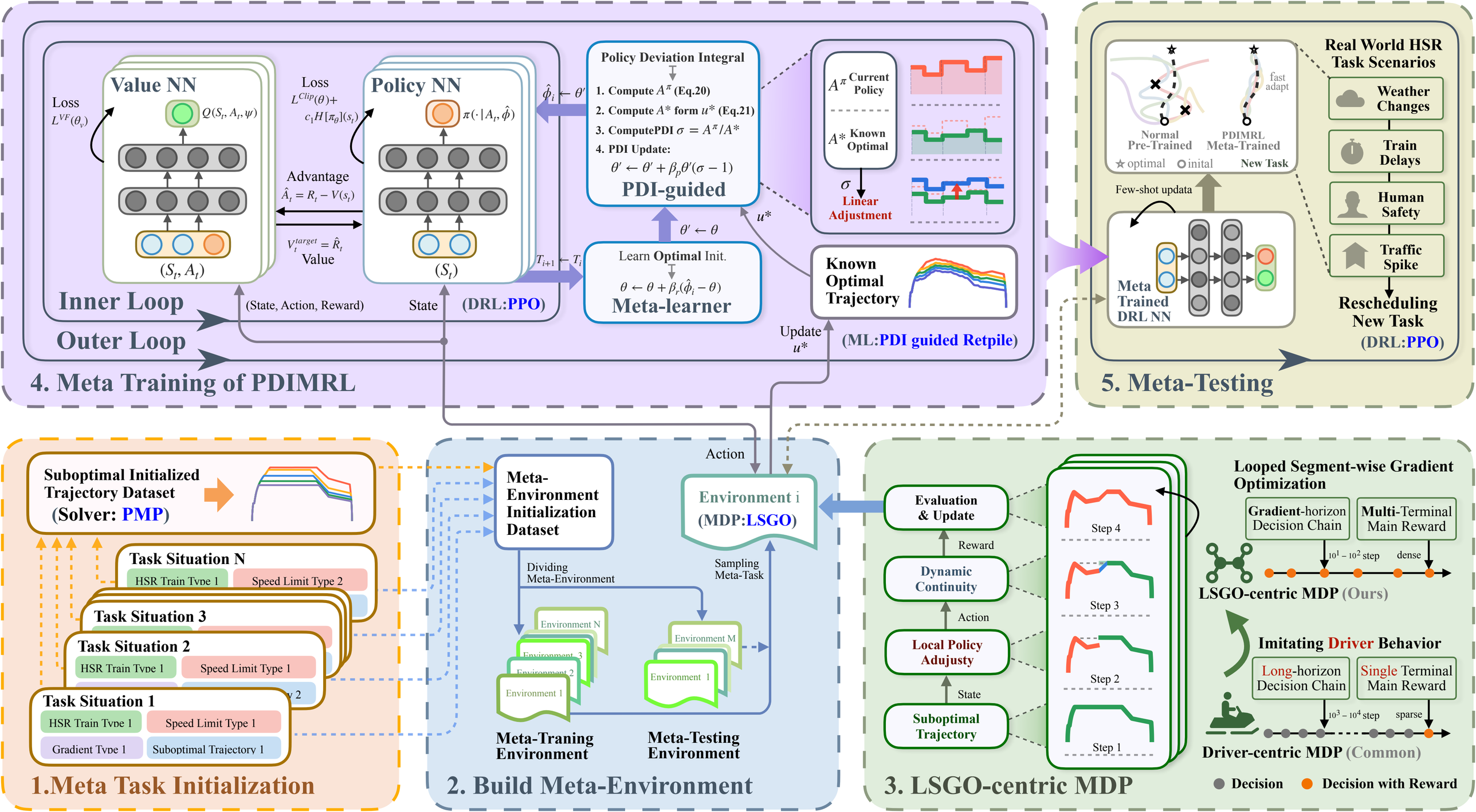
Value NN
Loss
$L^{VF}(\theta_v)$
$Q(S_t, A_t, \psi)$
(S_t, A_t)
Policy NN
Loss
$L^{Clip}(\theta)+$
$c_1 H[\pi_\theta](s_t)$
$\pi(\cdot|A_t, \hat{\phi})$
(S_t)
Advantage
$\hat{A}_t = R_t - V(s_t)$
$V_t^{target} = \hat{R}_t$
Value
Inner Loop
Outer Loop
(State, Action, Reward)
State
(DRL:PPO)
$\hat{\phi}_i \leftarrow \theta'$
$T_{i+1} \leftarrow T_i$
Policy Deviation Integral
1. Compute A^π (Eq.20)
2. Compute A^* form u^* (Eq.21)
3. ComputePDI $\sigma = A^\pi / A^*$
4. PDI Update:
$\theta' \leftarrow \theta' + \beta_p \theta'(\sigma - 1)$
PDI-guided
$\theta' \leftarrow \theta$
Learn Optimal Init.
$\theta \leftarrow \theta + \beta_r(\hat{\phi}_i - \theta)$
Meta-learner
u^*
A^π Current Policy
A^* Known Optimal
σ
Linear Adjustment
Known Optimal Trajectory
Update u^*
(ML:PDI guided Retpile)
4. Meta Training of PDIMRL
Real World HSR Task Scenarios
Weather Changes
Train Delays
Human Safety
Traffic Spike
Rescheduling New Task
(DRL:PPO)
Normal Pre-Trained
PDIMRL Meta-Trained
fast adapt
optimal
inital
New Task
Few-shot updata
Meta Trained DRL NN
5. Meta-Testing
Suboptimal Initialized Trajectory Dataset
(Solver: PMP)
Task Situation N
HSR Train Type 1
Speed Limit Type 2
Task Situation 3
Task Situation 2
HSR Train Type 1
Speed Limit Type 1
Task Situation 1
HSR Train Type 1
Speed Limit Type 1
Gradient Type 1
Suboptimal Trajectory 1
1.Meta Task Initialization
Meta-Environment Initialization Dataset
Action
Environment i
(MDP:LSGO)
Dividing Meta-Environment
Sampling Meta-Task
Environment N
Environment 3
Environment 2
Environment 1
Environment M
Environment 1
Meta-Traning Environment
Meta-Testing Environment
2. Build Meta-Environment
Evaluation & Update
Reward
Dynamic Continuity
Action
Local Policy Adujusty
State
Suboptimal Trajectory
Step 4
Step 3
Step 2
Step 1
3. LSGO-centric MDP
Looped Segment-wise Gradient Optimization
Gradient-horizon Decision Chain
Multi-Terminal Main Reward
$10^1 - 10^2$ step
dense
LSGO-centric MDP (Ours)
Imitating Driver Behavior
Long-horizon Decision Chain
Single Terminal Main Reward
$10^3 - 10^4$ step
sparse
Driver-centric MDP (Common)
Decision
Decision with Reward